Total No. of Questions : 8] [Total No. of Printed Pages : 3

Paper Code : 21321
F-421

B.C.A. (Fifth Semester)

(Main/Back) Examination, 2025-26

(New Course)

Paper - BCA-501-N

Computer Graphics And Animation

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any five questions. All questions carry equal marks. Symbols used have their usual meanings.

1. (a) Describe the development and role of computer graphics in modern computing.

   (b) Explain raster scan and random Sean display system with suitable diagrams.

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2. (a) Explain scan conversion. How is a line converted from mathematical from to raster form?

   (b) Write an algorithm for circle generation using the midpoint method.

3. (a) What is polygon filling? Explain any polygon filling technique.

   (b) Write short notes on:

   (i) Conic section generation

   (ii) Anti-aliasing

4. (a) Explain two-dimensional viewing. Discuss windowing and clipping operations.

   (b) Describe basic transformations in 2-D with examples.

5. (a) Explain interactive picture construction techniques in detail.

   (b) Write short notes

   (i) Interactive input devices

   (ii) Interactive output devices

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6. (a) Explain 3-D representation and transformations with suitable diagrams

(b) Describe the 3 -D viewing pipeline.

7. (a) What are spline curves and surfaces? Explain their applications.

(b) Explain any one algorithm for 3-D volume generation.

8. Write short notes on any four of the following

(a) Tweaking

(b) Morphing

(c) GKS Primitive

(d) Multimedia Applications

(e) Color display techniques

(f) Frame buffer

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